JEREMY WORSFOLD

CONTACT

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EXPERIENCE

POSTDOCTORAL RESEARCH FELLOW — 2024–present

School of Physics, UNSW Sydney | *Sydney, Australia*

- Stochastic processes and statistical physics in cell biology, with Richard Morris
- Developed a reduction of systems with fast multiplicative noise to switching processes on noise-induced manifolds; applied to autocatalytic reaction networks and epigenetic switching of histone marks
- Theory collaborator on two experimental projects: transcription dynamics, with Scott Berry and Bailey Hiles-Murison (designed the inference approach and wrote the statistical code), and force inference for T cells in collagen, with the Maté Biro lab (Garvan Institute); in both, modelling advice shaped experimental design

INNOVATION RESEARCH ASSISTANT — 2023–2024

University of Warwick | *Coventry, UK*

- Computational models of microtubule assembly, with Nigel Burroughs

EDUCATION

PhD in Mathematics: "Stochastic Active Flows" — 2020–2023

University of Bath | *SAMBa Centre for Doctoral Training*

(Awarded March 2024; supervisors: Tim Rogers and Paul Milewski)

- Developed a general model of randomly interacting particles and derived its finite-size effects analytically
- Discovered a novel form of synchronisation between oscillators driven by purely random forcing, and characterised its steady states and low-dimensional dynamics
- Showed with a stochastic model how random lane changes slow traffic

MRes in Statistical Applied Mathematics — 2019–2020

University of Bath | *SAMBa Centre for Doctoral Training*

- First class; modules included applied SDEs, advanced dynamical systems and scientific computing
- Group projects: statistical inference of MRSA transmission networks between hospitals; reinforcement learning for aircraft navigation around ash clouds, with Rolls-Royce

MPhys in Theoretical Physics — 2015–2019

University of Manchester | *UK*

- First Class, with Special Commendation (average above 80%)
- Master's dissertation: "Analysing Spatial Control in Football", in collaboration with Manchester City Football Club

PUBLICATIONS

J. Worsfold and R. G. Morris (2025). "Stochastically bistable growth and decay in the Togashi–Kaneko model". In: *Physical Review E* 112.1, p. 014131. DOI: 10.1103/yf5b-mljb.

J. Worsfold and T. Rogers (2024a). "Collective synchronization through noise cancellation". In: *Physical Review E* 109.2, p. 024218. DOI: 10.1103/PhysRevE.109.024218.

J. Worsfold and T. Rogers (2024b). "Stay in your lane: Density fluctuations in multi-lane traffic". In: *Europhysics Letters* 145.1, p. 11002. DOI: 10.1209/0295-5075/ad1c05.

J. Worsfold, T. Rogers, et al. (2023). "Density fluctuations in stochastic kinematic flows". In: *SIAM Journal on Applied Mathematics* 83.3, pp. 1000–1024. DOI: 10.1137/22M1494166.

Manuscripts in preparation

J. Worsfold[*], B. Hiles-Murison[*], et al. "Autocorrelation-based inference of transcription dynamics via infinite-server queueing models".

J. Worsfold, P. Ryan, et al. "Noise-induced manifolds: reduction of slow–fast SDEs to switching processes". Complete draft, available on request.

P. L. Newman[*], **J. Worsfold**[*], et al. "Fibre-resolved traction force microscopy".

P. Ryan, **J. Worsfold**, et al. "Functionally adaptive collective evasion via noise-induced switching".

[*]Equal contribution.

TALKS

Invited

- "Autocorrelation-based inference of transcription dynamics via infinite-server queueing models". Imperial College London (2026)
- "Density fluctuations in stochastic kinematic flows". University of St Andrews (2023)

Contributed

- "Autocorrelation-based inference of transcription dynamics via infinite-server queueing models"
  - SMB Annual Meeting, Graz (2026)
  - Sydney Mathematical Biology Symposium (2025)
- "How do T cells migrate through collagen? Inferring subcellular forces from fibre-resolved extracellular matrix deformations". SMB Annual Meeting, Graz (2026)
- "Stochastically bistable growth and decay in the Togashi–Kaneko model". ANZIAM (2025)
- "Collective synchronization through noise cancellation"
  - SIAM Conference on Applications of Dynamical Systems, Portland (2023)
  - BAMC, Bristol (2023)
- "Density fluctuations in stochastic kinematic flows"
  - BAMC, Loughborough (2022), **Best Student Talk**
  - Poster, SMB Annual Meeting, Heidelberg (2022)

TEACHING & SUPERVISION

GUEST LECTURER — 2026

University of New South Wales

- Guest lecture on reaction kinetics in *Principles of Biophysics*, a second-year course in the Biophysics major

RESEARCH MENTORING — 2024–present

University of New South Wales

- Mentored Patrick Ryan from Honours into his PhD: led meetings during his supervisor's absence, helped him develop the central idea of his first manuscript, and brought him into the noise-induced-manifolds project as a co-author

LATEX TUTORIAL — 2025–2026

UNSW Physics Society

- Optional tutorial for 30 students, from beginners to experienced users, on lab-report writing and formatting; run again in 2026 by invitation and asked back for a third year

UNDERGRADUATE TUTOR — 2019–2023

University of Bath

- Tutored 2nd-year *ODEs and Control* and *Modelling and Dynamical Systems* (2020–2023) and 1st-year MATLAB programming (2019–2020): groups of 20–30, tutorials, marking, worked examples on common misunderstandings
- Made Mathematica and Python visualisations and weekly lecture summaries to build intuition

PROGRAMMING TRAINING FOR NEW PHD STUDENTS — 2022–2023

University of Bath

- Restructured and ran the CDT programming workshop around transferable tools (git, coding style, LaTeX), with lecturers describing their own; the format was kept after I left

SERVICE & LEADERSHIP

Peer review

- Referee for Royal Society Open Science and PLOS Complex Systems

Organising

- Co-organiser, weekly postgraduate seminar series, Bath (2020–2021)
- Treasurer, Bath SIAM Student Chapter (2022–2023)
- Drafted the Morris Group (UNSW) code of conduct for a safe working environment
- Co-president, Just Love Manchester (2018–2019): led a committee running volunteering and fundraising events